ReliefChain Engine

Disaster-Relief Supply-Chain & Priority Queue Dispatch Visualizer

Simulation Ready | Reset System

Live Network Simulation | **Strategy Comparison & Benchmark** | Dynamic Max-Heap Inspection | C Engine & Algorithmic Docs

Multi-Strategy Comparative Benchmark

Simulating all 4 dispatch strategies on identical initial conditions with capacity constraints and dynamic events.

Re-Run Benchmark Comparison

BEST COMPOSITE SCORE	AVG RESPONSE TIME	JAIN'S FAIRNESS INDEX	DEMAND SATISFACTION
95.42	1.67 ticks	0.997	94.9%
Hybrid-Weighted Strategy	Fast First-Dispatch	Equitable Distribution	High Fulfilment

Quantitative Strategy Comparison Matrix

STRATEGY	AVG RESPONSE TIME	JAIN'S FAIRNESS INDEX	DEMAND SATISFACTION	SUPPLY UTILIZATION	STARVED ZONES	FINISH TICK	COMPOSITE SCORE
Urgency-First	1.67 ticks	0.983	90.7%	95.0%	0	20	93.70
Demand-First	2.50 ticks	0.995	95.0%	99.5%	0	20	93.95
Distance-First	4.33 ticks	0.985	93.8%	98.2%	0	20	90.02
Hybrid-Weighted	1.67 ticks	0.997	94.9%	99.3%	0	20	95.42

Urgency-First

Prioritizes zones with highest danger rating (1-10) plus aging bonus. High speed for critical emergencies, but can neglect large distant zones.

Score: --

Demand-First

Ranks zones by highest total volume of unmet demand. Highly efficient at clearing large inventory, but high urgency small zones may wait.

Score: --

Distance-First

Favors zones closest to active warehouses (`1000 / (dist + 1)`). Minimizes route delay, but remote critical zones can be starved.

Score: --

RECOMMENDED

Hybrid-Weighted

Multi-criteria optimization score: `0.45*Urgency + 0.25*Demand + 0.20*Distance + 0.10*Aging`. Delivers the optimal balance across response speed, fairness, and supply satisfaction!

Score: --